

HEMANATH T

Data Scientist | Machine Learning Engineer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [LeetCode](#)

PROFILE SUMMARY

Aspiring Data Scientist and Machine Learning Engineer with hands-on experience building machine learning, deep learning, computer vision, and analytics solutions. Skilled in Python, SQL, TensorFlow, Scikit-learn, computer vision, and model deployment. Developed end-to-end projects involving disease classification, object detection, clustering, recommendation systems, and AI-powered applications.

TECHNICAL SKILLS

Programming: Python, SQL

Data Analysis: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing, Feature Engineering, Statistical Analysis

Machine Learning: Regression, Classification, Clustering, PCA, Decision Trees, Random Forest, Ensemble Learning, Model Evaluation

Deep Learning: ANN, CNN, TensorFlow, Transfer Learning

Computer Vision: Object Detection, YOLO, RF-DETR, OpenCV

Visualization & Apps: Matplotlib, Seaborn, Plotly, Streamlit

Tools: Git, GitHub, FastAPI, Jupyter Notebook, Google Colab

PROJECTS

CropGuard AI — Tomato Leaf Disease Detection | TensorFlow, ResNet50, FastAPI, Gemini

- Built an end-to-end tomato leaf disease classification system using 14,529 images across 10 classes.
- Fine-tuned an ImageNet-pretrained ResNet50 using transfer learning and class-weighted training; evaluated performance on held-out and unseen real-world images.
- Developed and deployed a FastAPI inference application with confidence scores, top predictions, and Gemini-powered disease management recommendations.

AI-Based Rumble Strip Detection & Driver Warning System | Python, RF-DETR, Roboflow, OpenCV

- Built and annotated a custom 99-image highway dataset with positive and hard-negative road scenes; trained an RF-DETR Small object detection model.
- Achieved 90.1% mAP@50, 100% precision, 90% recall, and 94.7% F1 on external validation.
- Developed a video inference pipeline with confidence filtering and temporal confirmation, detecting 3/3 rumble-strip events in previously unseen highway footage and generating early driver warnings.

Football Player Analytics Dashboard | Python, Pandas, Scikit-learn, Plotly, Streamlit

- Analyzed 17K+ players, 768+ clubs, and 1.8M+ appearance records using data cleaning, EDA, and feature engineering.
- Applied StandardScaler, K-Means clustering, and PCA to segment players and visualize player profiles based on performance metrics.
- Built a Nearest Neighbors recommendation engine for similar-player discovery and deployed an interactive Streamlit dashboard.

Customer Churn Prediction | Python, Scikit-learn, XGBoost, SMOTE, Streamlit

- Built an end-to-end telecom customer churn prediction system using feature engineering, classification models, and SMOTE-based class balancing.
- Achieved 0.861 ROC-AUC and 0.791 recall, and deployed an interactive Streamlit application for churn-risk prediction.

EDUCATION

M.E COMPUTER SCIENCE AND ENGINEERING - CGPA OF 8.0 / 10

Hindusthan College of Engineering and Technology, Coimbatore | 2024-2026

B.E COMPUTER SCIENCE AND ENGINEERING - CGPA OF 8.6/10

Hindusthan College of Engineering and Technology, Coimbatore | 2019-2023

COURSEWORK

Data Science & Analytics Program — ProITbridge | Expected Completion: Sep 2026

Python, SQL, Statistics, Machine Learning, Deep Learning, Data Visualization, Power BI/Tableau, Model Deployment.